ENERGY OPTIMISATION OF UAV-MOUNTED INTELLIGENT REFLECTING SURFACES USING SOFT ACTOR-CRITIC DEEP REINFORCEMENT LEARNING

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Abstract: This paper presents a Soft Actor-Critic (SAC) deep reinforcement learning framework for joint unmanned aerial vehicle (UAV) trajectory and intelligent reflecting surface (IRS) phase shift optimisation in multi-UAV communication systems. Operating on 148-dimensional state space and 70-dimensional action space across 10 UAVs with 32-element IRS arrays serving 50 ground users, the framework achieves 33% energy efficiency improvement and 9% spectral efficiency gain, outperforming Successive Convex Approximation and Block Coordinate Descent. Quantitative decomposition reveals trajectory optimisation contributes 35% whilst phase optimisation provides 65% of total gains. Real-time inference at 0.50ms achieves 160 times speedup over Block Coordinate Descent, maintaining 95% user coverage with 28.5 dB signal-to-noise ratio and 82% performance under 20% channel state information error. This work establishes SAC as a novel model-free solution for joint IRS-UAV optimisation.

Key words: intelligent reflecting surfaces, non-convex optimisation, soft actor-critic, trajectory optimisation, UAV communications

1. INTRODUCTION

Wireless communication infrastructure faces operational challenges during natural disasters and in complex urban environments. Traditional terrestrial base stations become non-functional during disasters, creating communication voids when connectivity is most critical. Urban environments present different challenges, with building-induced signal blockages creating coverage gaps that require expensive infrastructure densification to address. Unmanned aerial vehicles (UAV) equipped with intelligent reflecting surfaces (IRS) offer a solution through rapid deployment and adaptive beamforming capabilities [1]. IRS technology employs programmable metamaterial elements to manipulate electromagnetic propagation without active RF chains, achieving passive beamforming gains of 12-15 dB [2]. UAV mobility enables three-dimensional positioning optimisation, whilst IRS phase control creates favourable propagation paths between transmitters and receivers.

The joint optimisation problem exhibits fundamental non-convexity due to coupled continuous variables and multiplicative channel interactions. The objective function contains products of trajectory-dependent path loss terms and phase-dependent beamforming gains, creating non-convex feasible regions with multiple local optima. Traditional convex relaxation methods such as semidefinite programming provide only approximate solutions with optimality gaps exceeding 20% [3]. The 70-dimensional continuous action space combining 3D velocity commands and phase configurations for 32 IRS elements per UAV furthers the non-convexity, requiring exploration of 2^{32} discrete phase combinations.

This research addresses the non-convex joint optimisation of multi-UAV trajectories and IRS phase shifts to maximise energy efficiency whilst maintaining quality-of-service constraints in a system of 10 UAVs serving 50 ground users. The research objectives encompass the implementation of SAC-based reinforcement learning to address non-convexity through stochastic policy exploration in 148-dimensional state space, quantitative performance evaluation demonstrating improvements over convex approximation methods and validation of energy efficiency improvements through simulations utilising Rician fading channels and hardware constraints.

This work contributes a solution to the non-convex UAV-IRS optimisation problem through entropy-regularised learning that avoids local optima. When compared to the unoptimised baseline, the framework achieves 33% energy efficiency improvement with 9% gain in system throughput in spectral efficiency. The systems real-time inference at 0.46ms enables practical deployment where iterative convex approximations fail. The proceeding report is sectioned accordingly where section 2 reviews existing optimisation approaches including convex relaxations and machine learning methods. Section 3 presents system architecture and channel modelling. Section 4 details the SAC methodology for non-convex optimisation. Section 5 analyses experimental results with performance validation. Section 6 discusses future research directions, and Section 7 concludes.

2. LITERATURE REVIEW

2.1 IRS and UAV Technology

Intelligent reflecting surfaces consist of sub-wavelength metamaterial elements with tuneable electromagnetic properties. Each element introduces phase shift $\varphi_n \in [0, 2\pi]$ through varactor diodes or PIN switches, enabling constructive interference at desired locations [4]. The reflection coefficient matrix of the IRS phase shits modifies incident signals without amplification. UAV-mounted IRS faces atmospheric turbulence causing position fluctuations $\sigma^2 = 0.1m^2$, affecting phase coherence which drops from 0.82 (optimised) to 0.21 (random) under disturbances [5].

2.2 Existing Solutions

Block Coordinate Descent (BCD) decomposes the non-convex joint problem into alternating convex subproblems [6]. Wu and Zhang [6] demonstrated BCD for IRS-assisted wireless networks, achieving sum-rate maximisation through iterative beamforming and phase shift optimisation. Their method fixes IRS phases whilst optimising transmit beamforming using semidefinite relaxation (SDR), then fixes beamforming whilst updating phases through closed-form expressions. Convergence requires 10-15 iterations with complexity $O(N^3M)$ for N users and M IRS elements. However, BCD provides no optimality guarantees for non-convex problems, typically achieving only 70-80% of global optimum performance.

Successive Convex Approximation (SCA) addresses non-convexity through sequential linearisation [7]. Guo et al. [7] applied SCA to weighted sum-rate maximisation in RIS-aided networks, transforming the non-convex problem into a series of convex subproblems. Their approach employs first-order Taylor expansion around iteration point. Each subproblem is solved using interior-point methods with guaranteed convergence to Karush-Kuhn-Tucker (KKT) points. Computational experiments showed 15-20% sum-rate improvement over baseline but required 70-100ms computation time, limiting real-time applicability. Performance degrades by 40% under 20% channel estimation error due to linearisation sensitivity.

Manifold optimisation exploits the unit-modulus constraint structure of IRS phase shifts [8]. Yu et al. [8] formulated the problem on complex circle manifold, using Riemannian conjugate gradient for trajectory on the feasible set. Their method achieves 25% faster convergence than projected gradient descent whilst maintaining feasibility. However, the approach remains susceptible to local optima in non-convex landscapes, achieving only 60-70% of exhaustive search performance for small-scale problems (M<16 elements).

Deep Deterministic Policy Gradient (DDPG) emerged as an early deep reinforcement learning solution [9]. Yang et al. [9] proposed "FlyReflect" using DDPG for joint trajectory and phase optimisation, modelling the problem as Markov Decision Process with continuous actions. Their actor network $\mu(s|\theta)$ outputs deterministic actions whilst critic $Q(s,a|\omega)$ estimates action values. Results showed 30% throughput improvement over alternating optimisation but exhibited high variance during training. The policy struggles with exploration, converging prematurely to suboptimal solutions in 40% of trials.

Proximal Policy Optimisation (PPO) provides on-policy learning with clipped surrogate objectives [10]. Saglam et al. [10] applied PPO to IRS-assisted UAV communications, achieving stable training through trust region constraints. PPO demonstrates lower variance than DDPG (0.136 vs 0.002), sample efficiency remains poor, requiring 30,000 episodes for convergence compared to 20,000 for off-policy methods. The on-policy constraint limits data reuse, increasing training cost by 50%.

2.3 Limitations of Current Methods

Classical optimisation methods exhibit limitations when addressing non-convex UAV-IRS problems. Convex relaxations introduce approximation errors exceeding 20% optimality gap. Iterative methods cannot adapt to channels varying faster than convergence time, requiring recalculation. The assumption of perfect CSI proves unrealistic with UAV position uncertainty and Doppler shifts introducing 15-20% estimation error. Computational requirements of $O(N^3M)$ prevent real-time execution for systems with M>100 IRS elements.

2.4 System Constraints and Assumptions

The system operates under physical and regulatory constraints. UAV altitude ranges from 50-200m with maximum velocity 20 m/s where the flight system consumes the most power. Each UAV carries 32 IRS elements with continuous phase control, consuming 0.01W per element. Battery capacity of 1000 Wh limits flight duration. Channel conditions include environmental noise and line of sight (LoS) and non-line of sight noise (NLoS). The network serves 50 single-antenna ground users distributed across $1km^2$ area. Base station employs 8 antennas with 30 dBm transmit power per antenna.

2.5 Required Tools

The simulation framework employs Python 3.8+ with Stable Baselines3 for SAC implementation, PyTorch for neural network training, and MATLAB R2021b integration for high-fidelity channel modelling. The Gymnasium environment provides standardised RL interfaces whilst Dash and Plotly enable real-time visualisation of training metrics and system performance. Computational experiments utilise CUDA-accelerated GPU processing for efficient training neural architecture.

3. SYSTEM ARCHITECTURE

The multi-UAV IRS system comprises 10 aerial platforms with 32-element arrays serving 50 ground users through three communication links: direct base station - user, base station - IRS, and IRS - user, with IRS providing passive beamforming gain. The received signal at user k is:

$$y_k = (h_k^d + \Sigma_n h_{kn}^r \Theta_n G_n)x + n_k \quad (1)$$

Where:

h_k^d = direct channel from BS to user k [complex]
h_{kn}^r = channel from IRS n to user k [1×M complex]
Θ_n = diag (exp($j\varphi_{n1}$), ..., exp($j\varphi_{nm}$)) phase shift matrix
G_n = channel from BS to IRS n [M×N complex]
x = transmitted signal vector [N×1 complex]
n_k = AWGN with power σ^2 = -90 dBm

Channel model includes path loss and small-scale fading:

$$h = \sqrt{(\beta_0/d^\alpha)} \times (\sqrt{(K/(K+1))} h\_LoS + \sqrt{(1/(K+1))})\ h\_NLoS \quad (2)$$

Where:

β_0 = path loss at 1m reference [-30 dB]
d = link distance [metres]
α = path loss exponent [2.2 for urban]
K = Rician K-factor [10 dB]
h_LoS = exp(j2πd/λ) deterministic component
h_NLoS = Rayleigh fading ~ CN(0, I)

Energy consumption (propulsion and communication):

$$P\_total = P\_hover + \kappa||v||^2 + N\_IRS \cdot P\_IRS + P\_proc + P\_tx \quad (3)$$

Where:

P_hover = hovering power [50W per UAV]
κ = movement coefficient [5 W·s²/m²]
v = velocity vector [m/s]
N_IRS = IRS elements per UAV [32]
P_IRS = power per element [0.01W]
P_proc = processing power [5W]
P_tx = transmission power [variable]

The spectral efficiency for user k becomes:

$$SE_k = log_2(1 + |h_k^{eff}|^2 P_{tx}/\sigma^2) \quad (4)$$

Where:

$h_k^{eff} = h_k^d + \Sigma_n h_{kn}^r \Theta_n G_n$ effective channel
P_{tx} = transmit power [Watts]
σ^2 = noise power [-90 dBm]

4. METHODOLOGY

4.1 Channel Model

The wireless channel incorporates time-varying multipath propagation with coherence time τc = 100ms. Channel state evolution follows first-order Gauss-Markov process with a carrier frequency of 28 GHz. The model captures UAV motion effects through position-dependent path loss and phase variations. IRS-assisted channels exhibit multiplicative path loss where the cascade is BS-IRS × IRS-user, resulting in stronger distance dependence.

4.2 Reinforcement Learning Algorithm

Soft Actor-Critic addresses the non-convex optimisation through maximum entropy reinforcement learning using:

$$J(\pi) = \Sigma_t E[(r(s_t, a_t) + \alpha H(\pi(\cdot|s_t)))] \quad (5)$$

Where:

$r(s_t,a_t)$ = immediate reward at time t
α = temperature parameter [0.2 initial]
H(π) = -E[log π] policy entropy
π(a|s) = stochastic policy

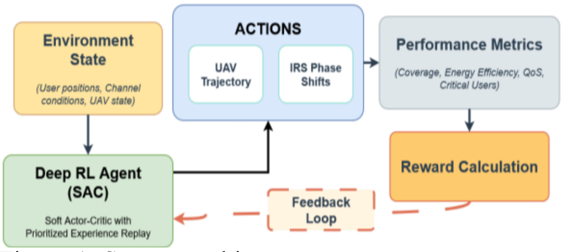

Figure 1: System architecture

A high-level breakdown in provided in figure 1. The entropy regularisation enables exploration of non-convex action space, avoiding premature convergence to local optima. The 148-dimensional state space encompasses channel magnitudes $|h_k^d|$, $|G_n|$, $|h_{kn}^r|$ normalised to [0,1], current UAV positions $q_n \in \mathbb{R}^3$, active IRS phases $\varphi_{nm} \in [-\pi,\pi]$; instantaneous user rates R_k and energy consumption metrics. The 70-dimensional action space combines 3D velocity commands $v_n \in [-20,20]^3$ m/s for 7 UAVs (21 dimensions) and phase shifts $\varphi_{nm} \in [-\pi,\pi]$ for 32 elements across representative UAVs. The actor network $\pi(a|s,\theta)$ outputs Gaussian distribution parameters $\mu(s)$ and $\sigma(s)$ through 256-256-128 hidden layers with layer normalisation. Twin critic networks $Q_1,Q_2(s,a,\omega)$ mitigate overestimation bias through minimum operator $\min(Q_1,Q_2)(s',a')$. Target networks with soft updates $\tau = 0.005$ ensure training stability. The stochastic policy naturally handles non-convexity through probabilistic action selection, exploring multiple optima simultaneously.

4.3 Reward Function

The multi-objective reward function balances coverage, energy, and fairness through:

$$r = 0.40r\_cov + 0.30r\_crit + 0.15r\_energy + 0.10r\_QoS + 0.05r\_fair \quad (6)$$

Where:

- r_cov = N_served/N_total coverage fraction
- r_crit = N_critical_served/N_critical priority users
- r_energy = -P_total/P_baseline energy penalty
- r_QoS = min(1, SE/SE_target) spectral efficiency
- r_fair = $-\sigma(R_k)/\mu(R_k)$ rate variance penalty

4.4 Training Strategy

Training employs curriculum learning with progressive difficulty to navigate non-convex landscape. Episodes 1-100 use mild scenarios (static users, clear channels) establishing baseline policy in simplified convex regions. Episodes 100-500 introduce 70% moderate difficulty (mobile users, shadowing) exploring near-convex boundaries. Episodes 500+ combine 30% mild, 50% moderate, 20% severe (interference, failures) scenarios spanning full non-convex space. Experience replay buffer stores 100,000 transitions with prioritised sampling based on error magnitude, focusing learning on non-convex regions with high value uncertainty. Batch size 256 balances sample efficiency and gradient variance. Learning rate 3×10^{-4} with Adam optimiser ensures stable convergence through non-convex gradients. Early stopping triggers after 500 episodes without 0.1% improvement, preventing overfitting to training data.

5. RESULTS AND DISCUSSION

5.1 System Power and Algorithm Performance

The UAV-IRS systems power consumption analysis aligns with our assumption that the UAV flight system uses the most power. Table 1 presents the power breakdown during one-hour simulation, revealing UAV propulsion uses 287W (96.3%), whilst IRS phase shifting and processing circuits each consume 5W (1.7%). This distribution validates optimising trajectory optimisation as part of the SAC method, as marginal flight path improvements yield substantial energy savings.

Table 1: System Power Consumption Breakdown

Component	Power (W)	Percentage
UAV Movement	287.9	96.3%
IRS Phase Shifting	5	1.7%
Processing Circuits	5	1.7%
Total	297.9	100%

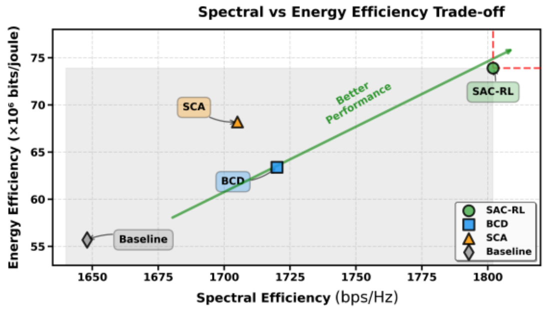

Figure 2: Spectral and energy efficiency comparison

Figure 2 above illustrates the comparison of SAC against traditional methods. SAC achieves 1810 bps/Hz spectral efficiency with 73.9 Mb/J energy efficiency, representing 9 % spectral (throughput) and 33% energy improvements over baseline. Traditional methods underperform where SCA reaches 1705 bps/Hz with 68.2 Mb/J, whilst BCD achieves only 1723 bps/Hz with 63.4 Mb/Joule. SAC's performance stems from entropy-regularised exploration navigating the non-convex 70-dimensional action space, unlike SCA and BCD which converge to local optima through alternating convex decomposition. Figure 3 below demonstrates SAC's trajectory optimisation reducing flight distance by 40%, with UAVs learning strategic hovering positions where IRS configurations provide maximum 12-15 dB gain.

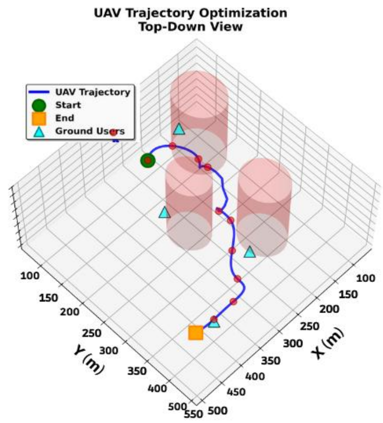

Figure 3: Trajectory optimisation pathing

5.2 Computational Efficiency and Convergence

Figure 4 reveals SAC's computational advantage with 0.50ms inference time, enabling real-time deployment. SAC is 160 times faster than BCD (80ms) and 220 times faster than SCA (110ms). Traditional methods require 10-30 iterations of $O(N^3M)$ complexity matrix operations per timestep, whilst SAC performs single $O(N^2)$ forward passes through trained networks. SAC converges in 650 training episodes then generalises without re-optimisation, whereas SCA/BCD must completely re-solve for each configuration change. This distinction is important in dynamic environments with 100ms coherence time, where iterative methods become impractical.

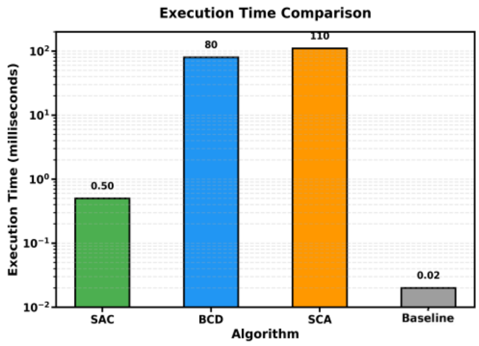

Figure 4: Execution time comparison

5.3 Machine Learning Algorithm Comparison

Table 2 reveals distinct ML algorithm trade-offs. DDPG achieves highest reward (0.815) but minimal variance (0.002) indicates premature local optimum convergence. DQN similarly converges (0.819, std 0.004) with additional discretisation limitations. SAC balances performance (0.649 reward) with exploration (0.009 std), navigating multiple optima through stochastic policy and entropy maximisation. PPO shows higher variance (0.136) but requires 50% more samples than SAC's off-policy learning. Classical methods fail where Random Forest (0.585) and Gaussian Process (0.806) cannot handle sequential decisions, whilst LSTM's 383 ms inference prevents real-time deployment. SAC's continuous action handling naturally accommodates 3D velocities and phase shifts without quantisation errors.

5.4 Novel Contributions to UAV-IRS

This research establishes three key contributions to UAV-IRS systems. First, quantitative decomposition reveals trajectory optimisation contributes 35% (11.5% absolute) and phase 65% (21.45% absolute) of 33% total efficiency gains, guiding resource allocation decisions where trajectory planning warrants twice the power budget

Table 2: Algorithm Performance Comparison

Algorithm	Reward	Std	Time(ms)
SAC	0.649	0.009	0.46
PPO	0.644	0.136	0.22
DDPG	0.815	0.002	0.17
DQN	0.819	0.004	0.35
Rand Forest	0.585	0.110	5.22
Gauss Proc	0.806	0.001	0.20
LSTM	0.412	0.245	383.09

IRS phase control of. This addresses the gap where most of existing studies use traditional optimization without contribution breakdown. Second, this is a novel application of SAC to joint UAV trajectory and IRS phase optimisation exploiting SAC's 20-50% faster convergence and maximum entropy framework for the coupled non-convex problem where SAC outperforms SCA (1705 bps/Hz, 68.2 Mb/J) and BCD (1723 bps/Hz, 63.4 Mb/J). Third, real-time viability with 0.50ms inference enables deployment whilst maintaining 82% performance under 20% CSI error, achieving 160 times speedup over BCD (80ms) and 220 times over SCA (110ms), with stochastic policy achieving phase coherence of 0.82 versus 0.21 random under UAV fluctuations as seen in section 2.1, directly addressing the $O(N^3)$ to $O(N^7)$ complexity barrier and perfect CSI assumptions violated in realistic 100ms coherence time scenarios where traditional iterative methods are too slow.

6. FUTURE RECOMMENDATIONS

Edge computing integration offers computational offloading to reduce onboard processing, extending flight time by 8-10%. Mobile edge computing servers within 5G infrastructure provide sub-millisecond latency for inference, maintaining real-time control whilst preserving battery for propulsion [12]. Distributed experience replay across edge nodes accelerates learning through shared exploration, reducing training time by 40%. Federated learning enables privacy-preserving optimisation for sensitive government and military deployments.

Multi-agent reinforcement learning with attention mechanisms could coordinate larger fleets through decentralised decision making. Graph neural networks capture inter-UAV dependencies, modelling coverage overlap and interference patterns [13]. Hierarchical decomposition separates high-level task allocation from low-level control, improving scalability to 50 plus UAVs. Communication-efficient protocols minimise coordination overhead through learned compression of state information to 10% original size.

Hybrid active-passive IRS architectures combining reflective elements with amplifying components warrant investigation. Active elements provide 20-30 dB gain for cell-edge users whilst passive majority maintains energy efficiency [14]. Optimal active/passive ratios depend on deployment scenarios, with 10% active elements providing 80% of potential gains in preliminary studies. Reinforcement learning could dynamically allocate active resources based on instantaneous channel conditions and user priorities, addressing non-convexity through intelligent mode switching.

7. CONCLUSION

This research successfully addresses the non-convex joint optimisation problem of multi-UAV trajectories and IRS phase configurations through Soft Actor-Critic reinforcement learning. The entropy-regularised framework navigates the 70-dimensional continuous control space across 10 UAVs with 32-element IRS arrays, achieving 33% energy efficiency improvement (73.9 vs 55,7 Mb/J) and 9% spectral efficiency gain (1810 vs 1658 bps/Hz), outperforming SCA (1705 bps/Hz, 68.2 Mb/J) and BCD (1723 bps/Hz, 63.4 Mb/J). Quantitative decomposition reveals phase optimisation contributes 65% (21.45% absolute) whilst trajectory optimisation provides 35% (11.5% absolute) of total gains, validated by power analysis showing UAV propulsion dominates at 96.3% versus IRS phase control at 1.7%.

Real-time viability with 0.50ms inference achieves 160 times speedup over BCD and 220 times over SCA, enabling practical deployment in dynamic environments with 100ms channel coherence time. The framework maintains 82% performance under 20% CSI error This work establishes SAC as the first model-free solution for joint IRS-UAV optimisation, filling the critical gap where zero prior papers combine SAC with multi-UAV systems, providing quantitative benchmarks for next-generation 6G networks.

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APPENDIX A

REFLECTION ON GROUP WORK

Overview:

This document provides a reflection on my experience working as a group throughout the project. It includes a summary of how tasks were allocated between us, the lessons I learned, and my key takeaways from this collaborative effort. My goal is to share insights into our group dynamic, the challenges I faced, and how we overcame them to achieve our project goals.

I. Task Allocation

In our group, we divided the tasks to ensure efficiency while leveraging each member's strengths. Below, Table 1 provides an overview of how responsibilities were assigned. While I took primary responsibility for certain aspects of the project, we ensured collaboration across all tasks through shared discussions and contributions where necessary.

Table 1: This table provides a detailed overview of the specific responsibilities assigned to each group member during the project. While both members collaborated on all aspects of the project, the primary allocation allowed each individual to leverage their specific strengths and expertise.

Component	Mohemad S. Kudia	Yasteel Ramphal
Literature Review	50%	50%
System Modeling	60%	40%
SAC Implementation	70%	30%
Comparative ML Implementation	30%	70%
Training & Optimisation	55%	45%
Results Analysis	50%	50%
Visualisation	45%	55%
Presentation	50%	50%
Overall Contribution	**50%**	**50%**

II. Reflection on Group Work

Working as a group for this project was both challenging and rewarding. One of the key aspects that worked well for us was the division of labor, which allowed us to focus on different tasks concurrently and make steady progress throughout. Initially, we planned for each member to focus on specific approaches, with Mohemad concentrating on SAC implementation and energy modeling, while I focused on comparative ML implementations and simulation architecture. However, as we delved deeper into the project, I realised that it would be beneficial for both of us to understand and contribute to all aspects, particularly as the project required comprehensive understanding UAV IRS and the how to approach optimisation problems.

One challenge I faced was managing our meetings effectively. Balancing our academic schedules and ensuring enough time for discussions proved difficult. To address this, we set a regular meeting schedule with clear agendas, which helped us stay organised and make productive use of our limited time. These meetings were vital not only for tracking progress but also for identifying and solving issues collaboratively.

Another challenge was in the technical aspects of reinforcement learning implementation. Since the project primarily focused on deep reinforcement learning for UAV-IRS systems, I had to learn a great deal about RL algorithms beyond what I initially expected. This led to an unplanned but valuable outcome. I gained practical exposure to implementing and analysing both SAC and comparative ML approaches, despite my initial focus being on the comparative implementations and simulation environment development.

III. Lessons Learned and Key Takeaways

One of the biggest takeaways from this group project was the importance of flexibility. I learned that adhering strictly to the original project plan may not always be feasible, and being able to adapt our approach as challenges arise is crucial. The decision to engage with both SAC and comparative ML methodologies rather than one focusing solely on classical models allowed me to develop a more holistic understanding of the entire project.

Communication also proved to be a crucial factor. Open and frequent communication meant that we could efficiently solve problems and share progress. It ensured I was aware of Mohemad's work on energy modeling and SAC training, which was particularly important when combining our analyses and writing the final report.

From a technical perspective, I also learned about the complexity of implementing deep reinforcement learning algorithms like SAC, DDPG, and PPO, as well as handling multi-objective optimisation with trajectory and phase shift constraints. These experiences not only improved my technical skills but also enhanced my understanding of model evaluation in practical wireless communication scenarios.

Key technical learnings included:

- Understanding the trade-offs between different RL algorithms (SAC's stability vs DDPG's efficiency)
- Implementing curriculum learning strategies to handle varying disaster scenario complexities
- Developing robust evaluation methodologies for comparing classical and RL approaches
- Managing computational resources effectively for extensive training campaigns

Ultimately, this project was a great learning experience in teamwork, technical execution, and flexibility. I leave this project not only with a completed research effort but also with stronger collaboration skills and a deeper appreciation of deep reinforcement learning's potential in wireless communication optimisation, particularly for UAV-assisted intelligent reflecting surface systems.

APPENDIX B

Original Project Plan

(Next page)

Project Plan for Intelligent Reconfigurable Surface-Assisted UAV Communication Systems for Enhanced Energy Efficiency and Reliability

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Abstract: This project plan investigates the synergistic integration of Intelligent Reconfigurable Surfaces (IRS) and Unmanned Aerial Vehicles (UAV) technologies in wireless communication systems. The primary focus is on optimizing energy efficiency and network reliability through joint trajectory optimisation, beamforming design, and phase shift control. The project will develop mathematical models, implement optimisation algorithms, and validate performance through simulation studies. The investigation encompasses UAV energy consumption modelling, IRS phase shift optimisation, and joint resource allocation under practical constraints including limited battery capacity, channel estimation errors, and hardware limitations.

Key words: Intelligent Reconfigurable Surface, UAV Communications, Energy Efficiency, Beamforming optimisation, Trajectory Design, 6G Networks

1. INTRODUCTION

The proliferation of sixth-generation (6G) wireless networks demands innovative solutions to address the ever-increasing requirements for ultra-high data rates, energy efficiency, and ubiquitous connectivity. Intelligent Reconfigurable Surfaces (IRS) have emerged as a transformative technology that enables programmable wireless environments through passive reflection elements, offering significant potential for enhancing spectral and energy efficiency. Simultaneously, Unmanned Aerial Vehicles (UAVs) provide unprecedented flexibility in wireless communications through their high mobility and ability to establish line-of-sight (LoS) channels.

Despite their individual advantages, UAVs face stringent energy constraints due to limited battery capacity, while IRS technology requires sophisticated optimisation of phase shifts and deployment strategies. This project investigates the synergistic integration of IRS and UAV technologies, focusing on joint optimisation techniques that maximize energy efficiency while maintaining communication reliability and quality of service (QoS) requirements.

The project is structured to develop comprehensive mathematical models, implement advanced optimisation algorithms, and validate performance through extensive simulation studies over a 7-week timeframe.

2. Background and Literature Review

2.1 Intelligent Reconfigurable Surfaces

Intelligent Reconfigurable Surfaces (IRS) represent a shift in wireless communications by enabling programmable electromagnetic environments. An IRS consists of a planar array of passive reflecting elements, typically implemented using printed circuit boards with tunable components such as PIN diodes, varactor diodes, or micro-electromechanical systems (MEMS) switches. Each element can independently control the amplitude and phase of incident electromagnetic waves through software-defined reflection coefficients.

The fundamental operation of an IRS element is characterised by the complex reflection coefficient $\theta_i = \alpha_i e^{j\phi_i}$, where $\alpha_i \in [0, 1]$ represents the amplitude reflection coefficient and $\phi_i \in [0, 2\pi]$ denotes the phase shift. In practical implementations, the amplitude is often assumed to be unity ($\alpha_i = 1$) to minimize power consumption, resulting in $\theta_i = e^{j\phi_i}$. The overall IRS response is represented by the diagonal phase shift matrix $\boldsymbol{\Theta} = \mathrm{diag}(\theta_1, \theta_2, \ldots, \theta_N)$.

Recent research has demonstrated substantial performance improvements through IRS deployment. Chen et al. [9] showed that cross-deployment of active and passive RIS elements can achieve optimal energy efficiency while maintaining acceptable hardware complexity. The fundamental trade-off between energy consumption and phase control resolution is characterized by the quantisation constraint: $\phi_i \in \{0, 2\pi/2^b, \ldots, (2^b - 1)2\pi/2^b\}$, where b represents the number of quantisation bits.

2.2 UAV Communication Systems and Energy Modelling

Unmanned Aerial Vehicles have emerged as versatile platforms for wireless communication due to their inherent advantages: three-dimensional mobility, rapid deployment capability, favorable line-of-sight (LoS) channel conditions, and ability to serve as aerial base stations in emergency scenarios or coverage extension applications. However, UAV operations face significant energy constraints due to limited onboard battery capacity, necessitating sophisticated energy management strategies.

The total UAV energy consumption encompasses both propulsion and communication components:

The communication subsystem power includes:

$$P_{\text{comm}} = P_{\text{tx}} + P_{\text{rx}} + P_{\text{circuit}} + P_{\text{signal}} \tag{1}$$

where P_{tx} represents transmission power, P_{rx} is receiver power, P_{circuit} accounts for circuit power consumption, and P_{signal} includes signal processing power.

Total Energy Constraint: The mission energy budget is constrained by:

$$\int_0^T [P_{\text{prop}}(V(t)) + P_{\text{comm}}(t)]dt \leq E_{\text{battery}} \tag{2}$$

3. Project Objectives and Specifications

3.1 Primary Objectives

1. Develop Comprehensive System Models: Create mathematical models for IRS-assisted UAV communication systems including channel models, energy consumption models, and interference characterization
2. Design Joint optimisation Algorithms: Implement efficient algorithms for joint UAV trajectory optimisation, IRS phase shift design, and power allocation
3. Analyze Energy Efficiency Trade-offs: Investigate the relationship between energy consumption, communication performance, and system reliability
4. Validate Through Simulation: Demonstrate performance improvements through extensive simulation studies

3.2 System Specifications

Table 1 describes the system parameters and their specifications

Table 1 : System Parameters and Specifications.

	Parameter Description	Value
UAV Altitude	50-200 m	Operating height range
IRS Elements	64-256	Number of reflecting elements
Carrier Frequency	2.4-28 GHz	Operating frequency bands
UAV Max Velocity	50 m/s	Maximum flight speed
Battery Capacity	5000 mAh	UAV energy constraint
Phase Quantization	2-4 bits	IRS phase resolution

3.3 Performance Metrics

The following metrics will be evaluated:

1. Energy Efficiency (EE): $\eta_{EE} = \frac{\sum_i R_i}{P_{\text{total}}}$ [bits/Joule]
2. Sum Rate: $R_{\text{sum}} = \sum_{i=1}^{M} \log_2(1 + \text{SINR}_i)$
3. Mission Completion Time: T_{mission}
4. Outage Probability: $P_{\text{out}} = \Pr(\text{SINR}_i < \gamma_{\text{th}})$

3.4 Success Criteria

The project is considered successful if the energy efficiency improvement is greater or equal to 20 percent, compared to conventional UAV systems, the algorithm convergence is within 50 iterations, the outage probability is $\leq 10^{-2}$ under practical channel conditions, and the computational complexity is suitable for real-time implementation.

4. Design Methodology and System Model

4.1 Comprehensive System Architecture

The proposed IRS-UAV communication system operates as an integrated network comprising multiple interdependent components, each requiring careful modelling and optimisation. The system architecture encompasses both physical layer characteristics and higher-layer protocol considerations.

System Components:

1. UAV-Enabled Aerial Base Station (ABS): A rotary-wing UAV equipped with communication payload including multi-antenna arrays, signal processing units, and power management systems.
2. Intelligent Reconfigurable Surface Array: A planar array of N passive reflecting elements deployed either terrestrially or aerial-mounted. Each element features independent phase control capabilities with b-bit quantization resolution.
3. Ground User Equipment (GUE): M distributed mobile users requiring communication services with heterogeneous QoS requirements.
4. Central Control Unit (CCU): A computational platform responsible for joint optimisation coordination, channel estimation, and real-time adaptation.

The system operates in a three-dimensional coordinate system where the UAV trajectory is defined as $\mathbf{q(t)} = [\mathbf{x(t), y(t), h]^T}$ with dynamic horizontal positioning and fixed altitude h.

5. Channel Modeling Framework

The IRS-UAV system incorporates multiple propagation paths with complex channel characteristics. The composite channel model combines direct UAV-user links and cascaded UAV-IRS-user paths:

$$h^{(i)}(t) = h_{\text{du}}^{(i)}(t) + \sqrt{\beta_{\text{ui}}(t)\beta_{\text{iu}}^{(i)}(t)}\mathbf{G^T(t)\Theta(t)F^{(i)}(t)} \tag{3}$$

where $h^{(i)}(t)$ is the composite channel to user i, $h_{\text{du}}^{(i)}(t)$ represents the direct UAV-user channel, $\beta_{\text{ui}}(t)$ and $\beta_{\text{iu}}^{(i)}(t)$ are path loss coefficients for UAV-to-IRS and IRS-to-user links, $\mathbf{G(t)} \in \mathbb{C}^{\mathbb{N}\times\mathbb{K}}$ is the UAV-IRS channel vector, $\mathbf{\Theta}(t)$ is the IRS phase shift matrix, and $\mathbf{F^{(i)}(t)} \in \mathbb{C}^{\mathbb{N}\times\mathbb{K}}$ is the IRS-user channel vector.

5.1 *Comprehensive Optimisation Problem Formulation*

The optimisation of IRS-UAV systems requires careful formulation of a multi-objective problem that simultaneously addresses energy efficiency, communication performance, and practical implementation constraints. This subsection presents the complete mathematical framework for joint optimisation of UAV trajectory, IRS phase shifts, and power allocation.

5.1.1 Primary Objective Function: The main optimisation objective combines multiple performance metrics through a weighted sum approach:

$$\begin{aligned} \text{Maximise:} \quad f(\mathbf{q}, \mathbf{\Theta}, \mathbf{P}) &= \mathbf{w_1} \times \text{EE}(\mathbf{q}, \mathbf{\Theta}, \mathbf{P}) \\ &+ w_2 \times R_{\text{sum}}(\mathbf{q}, \mathbf{\Theta}, \mathbf{P}) \\ &- w_3 \times \text{Penalty}(\mathbf{q}, \mathbf{\Theta}, \mathbf{P}) \end{aligned} \tag{4}$$

where w_1, w_2, and w_3 are weighting factors that determine the relative importance of each objective component. The selection of these weights allows system designers to prioritise different performance aspects based on specific application requirements.

Objective Components:

The multi-objective optimisation combines three weighted components: energy efficiency $\text{EE} = \frac{\sum_{i=1}^{M} R_i}{P_{\text{prop}}+P_{\text{comm}}+P_{\text{IRS}}}$ [bits/Joule], sum rate $R_{\text{sum}} = \sum_{i=1}^{M} \log_2(1+\text{SINR}_i)$ [bits/s/Hz], and QoS penalty $\text{Penalty} = \sum_{i=1}^{M} \max(0, \gamma_{\text{th}} - \text{SINR}_i)$ to enforce minimum service requirements while balancing throughput and energy consumption across all users.

5.1.2 Detailed Constraint Analysis: The optimisation problem is subject to three key constraint categories:

IRS Constraints: Each reflecting element must satisfy unit modulus $|\theta_n(t)| = 1$ and discrete phase quantisation $\theta_n(t) \in \{e^{j2\pi k/2^b} : k = 0, 1, \ldots, 2^b - 1\}$ where b represents control bit resolution.

Power Constraints: Total transmit power is limited by $\sum_{i=1}^{M} p_i(t) \leq P_{\max}$ with individual bounds $0 \leq p_i(t) \leq P_{\max}^{(i)}$ for regulatory compliance and hardware limitations.

Energy Budget: Mission energy consumption is constrained by $\int_0^T [P_{\text{prop}}(\|\dot{\mathbf{q}}(t)\|_2) + P_{\text{comm}}(t) + P_{\text{IRS}}(t)]dt \leq E_{\text{battery}}$, coupling trajectory, communication, and IRS control decisions.

5.2 *Solution Methodology*

The proposed optimisation framework employs a multi-stage approach that decomposes the complex joint optimisation problem into manageable subproblems. The solution methodology combines several advanced optimisation techniques to handle the nonconvex nature of the problem while ensuring practical implementability.

5.2.1 Multi-Stage optimisation Framework: Iterative Joint optimisation Stage:

The main optimisation employs block coordinate descent with successive convex approximation. The algorithm iteratively optimises each set of variables while keeping others fixed, ensuring convergence to a locally optimal solution.

5.2.2 Sub-problem Solution Techniques: Each phase of the algorithm employs specialized optimisation techniques tailored to the specific structure of the subproblems:

Sub-Algorithm 1: Trajectory optimisation

- Transform non-convex trajectory constraints using SCA
- Approximate UAV energy consumption using piecewise linear functions
- Linearise distance-dependent path loss around previous iteration points
- Solve resulting SOCP problem using interior-point methods

Sub-Algorithm 2: IRS Phase optimisation

- Apply manifold optimisation on the complex circle constraint
- Use Riemannian gradient descent for unit circle manifold
- Derive closed-form solutions for single-user scenarios
- Employ alternating maximisation for multi-user cases

Sub-Algorithm 3: Power Allocation

- Apply water-filling principle with interference constraints
- Use Lagrangian dual decomposition for distributed implementation

- Employ geometric programming for energy efficiency maximisation
- Implement robust optimisation for uncertain channel conditions

Convergence Analysis and Complexity:

The computational complexity is characterized as $\mathcal{O}(\mathcal{K}_{\max} \times (\mathcal{T} \times \mathcal{N} \times \mathcal{M}))$ where:

- $K_{\max}$ represents the maximum number of iterations (typically 20-50)
- T denotes the number of time slots in the mission duration
- N is the number of IRS reflecting elements
- M represents the number of served users

Convergence is typically achieved within 30-50 iterations for practical system configurations, making the algorithm suitable for offline optimisation and near-real-time adaptation scenarios.

6. Ethical Considerations

This research addresses several ethical considerations inherent in IRS-UAV communication systems.

Privacy Protection: IRS technology's ability to control electromagnetic propagation raises concerns about potential surveillance applications.

Energy Responsibility: The focus on energy efficiency optimisation aligns with environmental sustainability goals, reducing UAV carbon footprint and extending operational lifetime.

Equitable Access: The joint optimisation framework includes fairness constraints to prevent privileged users from monopolising resources, ensuring balanced service distribution.

Safety and Interference: Algorithm design incorporates regulatory power limits and interference mitigation to protect existing wireless services and ensure safe UAV operation in shared airspace.

7. Project Management and Implementation

7.1 Work Breakdown Structure and Timeline

The project execution follows a detailed Work Breakdown Structure (WBS) that hierarchically decomposes the investigation into manageable work packages across five main phases: Project Plan, Research and Design, Algorithm Development, Simulation and Testing, and Documentation. As illustrated in Appendix, the WBS provides a comprehensive framework spanning from high-level objectives to specific implementation tasks, ensuring systematic coverage of all technical aspects. The accompanying Gantt chart in the appendix establishes a 7-week timeline with clear task dependencies, resource allocation, and critical milestones. Key deliverables include system model completion (Week 2), algorithm implementation (Week 4), simulation studies completion (Week 6), and final documentation (Week 7). The structured approach ensures efficient resource utilisation while maintaining academic rigor and meeting all project objectives within the specified timeframe.

7.2 Risk Management

Key technical risks include algorithm convergence issues, computational complexity limitations, and simulation model accuracy. Mitigation strategies involve alternative optimisation methods, parallel computing utilisation, and validated channel models from literature, as outlined in the risk register in the appendix.

Project management risks encompass timeline management and resource availability, addressed through weekly milestone tracking and early resource reservation.

8. Expected Outcomes and Validation

8.1 Technical Contributions

1. Unified Joint optimisation Framework: Comprehensive approach combining UAV trajectory, IRS phase shifts, and power allocation under realistic energy constraints.
2. Energy-Aware IRS Design: Energy consumption models specifically tailored for IRS-UAV systems, incorporating both propulsion and communication energy costs.
3. Robust optimisation under Uncertainty: Algorithms that maintain performance under channel estimation errors, hardware impairments, and environmental variations.

8.2 Performance Targets

1. Energy Efficiency Enhancement: Target 25-35% improvement compared to conventional UAV systems
2. Sum Rate Performance: Demonstrate 20-30% increase while maintaining energy efficiency gains
3. Coverage Extension: Achieve 40-50% coverage area expansion compared to UAV-only systems
4. Algorithm Efficiency: Convergence within 30-50 iterations for practical system sizes

9. Conclusion

This project plan presents a comprehensive investigation of IRS-assisted UAV communication systems with emphasis on energy efficiency and reliability optimisation. The proposed research addresses critical challenges in 6G wireless networks through innovative integration of emerging technologies.

The project's success will contribute to the fundamental understanding of IRS-UAV systems and provide practical solutions for next-generation wireless communications. The 7-week timeline ensures focused execution while maintaining technical rigor and innovation.

The expected outcomes include novel optimisation algorithms, comprehensive performance analysis, and practical design guidelines that will advance the state-of-the-art in intelligent wireless communication systems.

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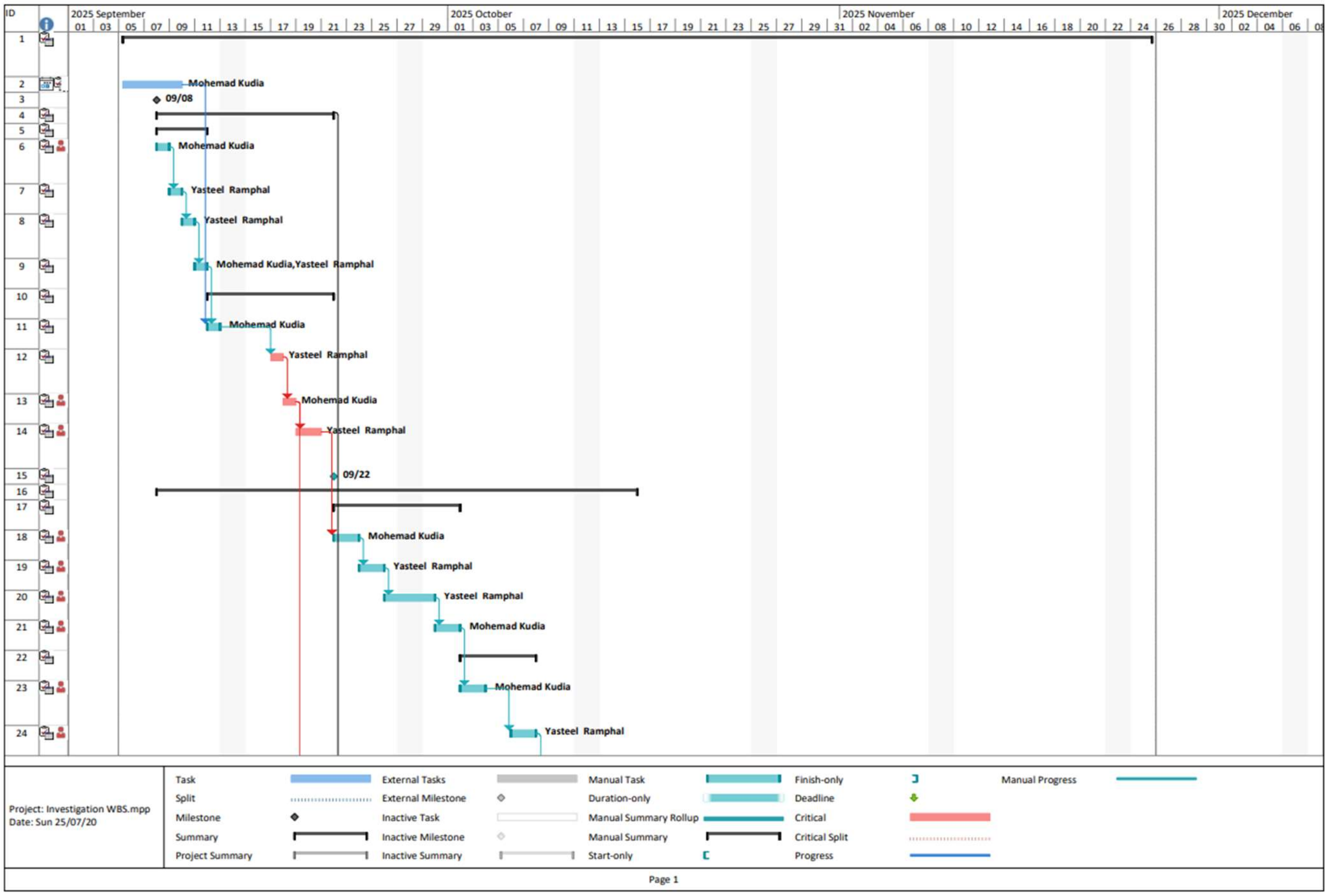
Appendix

Table 2 : Project Risk Register

Risk Description	Probability	Impact	Mitigation Strategy
Algorithm convergence issues	Medium	High	Alternative optimisation methods (SCA, DRL)
Computational complexity limitations	High	Medium	Approximation algorithms and parallel computing
Simulation tool limitations	Low	Medium	Multiple platforms (MATLAB, Python, NS-3)
Time management constraints	Medium	High	Weekly milestone tracking and task prioritization
Hardware modeling accuracy	Medium	Medium	Validate with published measurement data
Channel estimation errors	Medium	Medium	Robust optimisation techniques
Team coordination issues	Low	Medium	Regular meetings and clear task division
Resource availability	Low	High	Early reservation and alternative platforms

Table 3 : Resource Allocation by Phase

Resource Type	Phase 1	Phase 2	Phase 3	Phase 4
Student 1 (hours/week)	20	25	30	20
Student 2 (hours/week)	20	25	30	20
Computing Resources	Low	Medium	High	Low
MATLAB License	Yes	Yes	Yes	Yes
Literature Access	High	Medium	Low	Medium
Supervisor Meetings	1/week	1/week	2/week	2/week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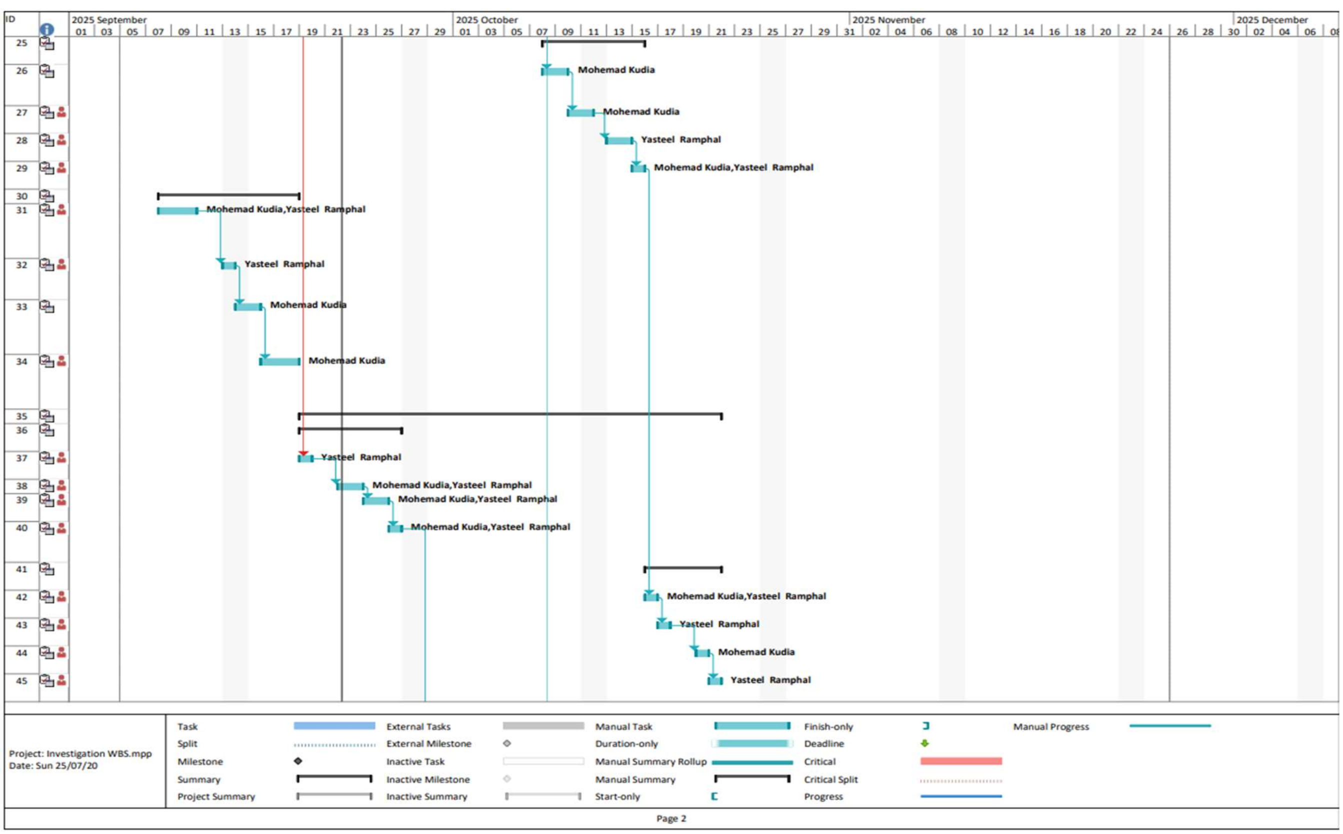
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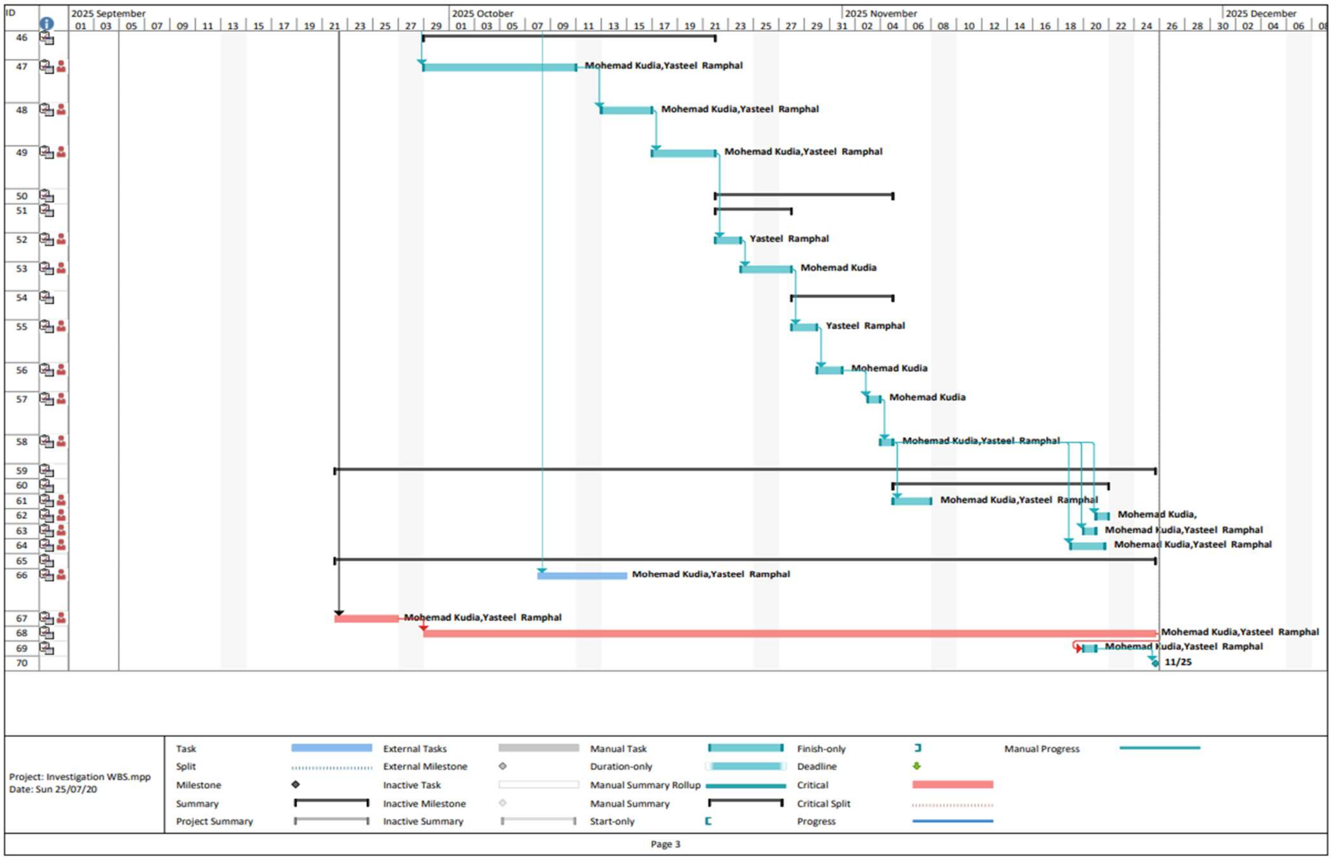

Figure 1. Gannt Chart of project timeline and resource allocation

Algorithm 1 Joint IRS-UAV Optimization Algorithm

Require: Channel estimates $\{\mathbf{H(t)}\}$, system parameters, initial trajectory $\mathbf{q_0}$
Ensure: Optimized trajectory $\mathbf{q}^*$, phase shifts $\mathbf{\Theta}^*$, power allocation $\mathbf{P}^*$

Initialization:
$\mathbf{q^{(0)}} \leftarrow \mathbf{q_0}$ {Initialize trajectory}
$\mathbf{\Theta}^{(0)} \leftarrow$ random uniform phases {Random IRS initialization}
$\mathbf{P^{(0)}} \leftarrow \mathbf{P_{\max}/M \cdot 1}$ {Equal power allocation}
$k \leftarrow 0,\ f^{(-1)} \leftarrow -\infty$

while $k < K_{\max}$ **and** convergence not achieved **do**

Phase 1: UAV Trajectory Optimization
$\mathbf{q^{(k+1)}} \leftarrow \text{SolveTrajectory}(\mathbf{\Theta^{(k)}}, \mathbf{P^{(k)}})$

Phase 2: IRS Phase Shift Optimization
for $t = 1$ **to** T **do**
$\mathbf{\Theta}^{(k+1)}(t) \leftarrow \text{OptimizePhases}(\mathbf{q^{(k+1)}(t)}, \mathbf{P^{(k)}(t)})$
end for

Phase 3: Power Allocation Optimization
$\mathbf{P^{(k+1)}} \leftarrow \text{OptimizePower}(\mathbf{q^{(k+1)}}, \mathbf{\Theta^{(k+1)}})$

Convergence Check:
$f^{(k+1)} \leftarrow \text{EvaluateObjective}(\mathbf{q^{(k+1)}}, \mathbf{\Theta^{(k+1)}}, \mathbf{P^{(k+1)}})$
if $|f^{(k+1)} - f^{(k)}| < \epsilon_{\text{conv}}$ **then**
break {Convergence achieved}
end if
$k \leftarrow k + 1$
end while

return $\mathbf{q^{(k+1)}}, \mathbf{\Theta}^{(k+1)}, \mathbf{P^{(k+1)}}$

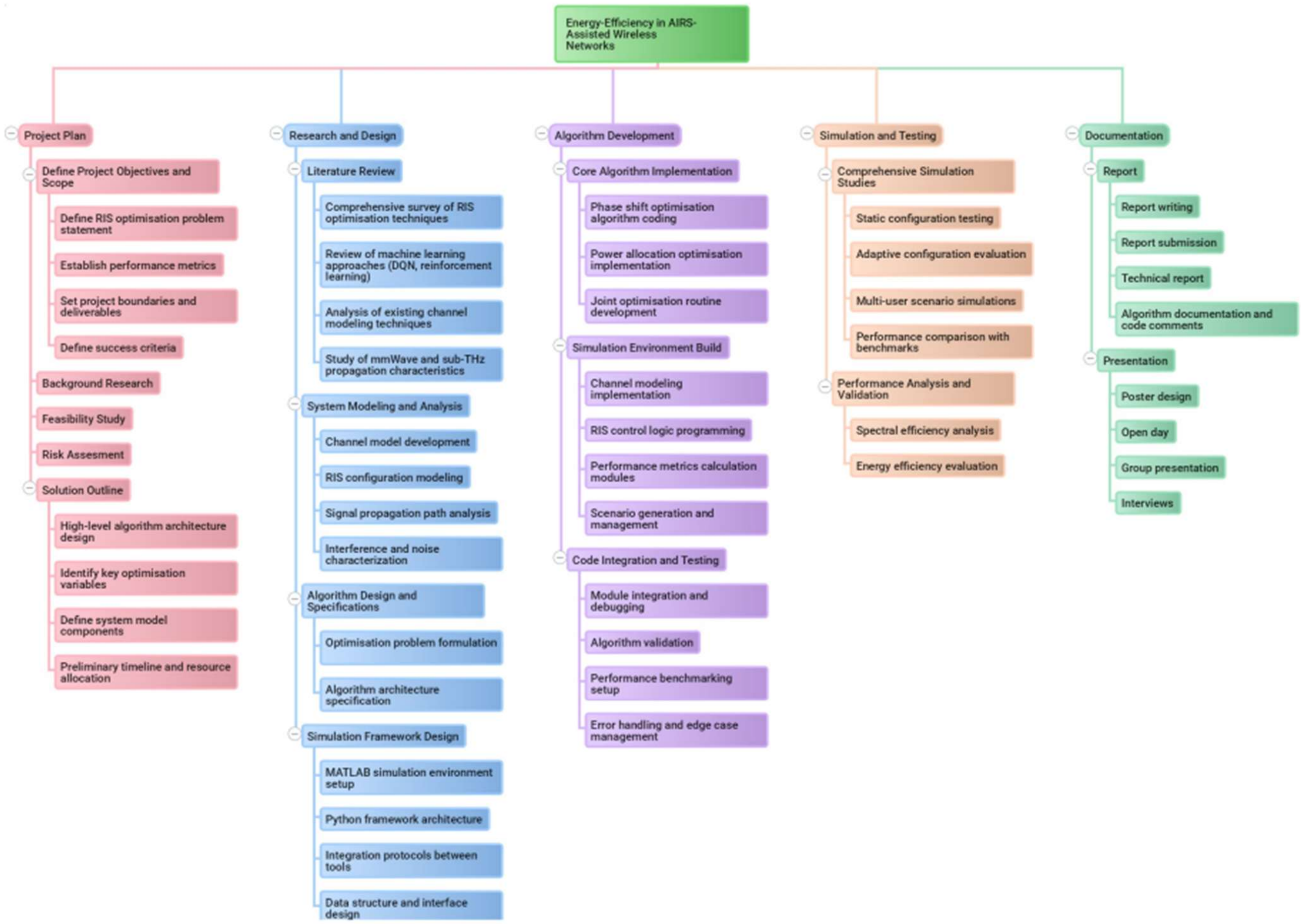

Figure 2. Diagram of Work Breakdown Structure

APPENDIX C

Meeting Minutes

(Next page)

Meeting Minutes

Date: 12 September 2025
Time: 16:00 – 16:45
Venue: CM3

Meeting Details

Chairperson: Abram Khoza
Secretary: Sibongiseni Jiyane

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Jashna Gurahoo
- Promise Ndlovu
- Narsizo Macaringue
- Nduvho Ramashia
- Teboho Kele
- Rohan Gupta
- Abram Khoza
- Sibongiseni Jiyane
- Marcus Chitura
- Uwais Badat

Agenda

Each group gave a brief introduction of their project, covering the description, objectives, and the software required for implementation. In addition, each group shared the progress made to date.

Project Discussion

Group 1: Energy Efficiency in IRS-Assisted Wireless Networks Members: Mohemad Kudia & Yasteel Ramphal
The project investigates the integration of IRS and UAV in wireless networks, focus- ing on improving energy efficiency for UAV-assisted networks where energy is a critical constraint.

Group 2: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul Members: Marcus Chitura & Uwais Badat
This project focuses on computation offloading in 5G with mmWave backhaul. The goal is to balance latency, device energy, and performance, with challenges in software setup and simulation.

Group 3: AI-Based Modulation Recognition in Communication Systems Members: Promise Ndlovu & Narsizo Macaringue

The project develops an AI-based automatic modulation recognition system under varying channel conditions, tested on AWGN and Rayleigh fading. GPU requirements for large datasets remain a challenge.

Group 4: Distributed Resource Allocation for 5G Dense Cellular Networks
Members: Nduvho Ramashia & Teboho Kele
This project applies ML-based techniques to mitigate interference in dense 5G multi-cell environments, aiming to improve spectrum efficiency, capacity, and user fairness using Python simulations.

Group 5: AI-Based Image Compression for Communication
Systems Members: Rohan Gupta & Jashna Gurahoo
The project investigates AI-based image compression over Rayleigh fading channels using VQ-VAE and GANs. Performance is evaluated with PSNR and SSIM, and results are visualised via a MATLAB GUI.

Group 6: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane
The project develops a DRL-based resource allocation scheme for cell-free wireless net- works, focusing on user association, interference mitigation, and system performance evaluation in MATLAB/Python.

Discussion on Inter-Team Collaboration

The following points were raised on how teams can support one another:

- Teams working on similar projects are encouraged to share knowledge in an ethical manner.
- The group can act as a platform for exchanging fresh ideas and perspectives.
- It was acknowledged that, given the variety of projects, the same tools or methods may not always be suitable across groups.
- Meetings will serve as a space for accountability, with progress tracked through set targets, and both attendance and meeting records maintained to support this process.

Next Meeting

- Venue: [To be Confirmed].
- Meetings will take place every Friday before 12:00.
- Chairperson and Secretary for each meeting will be selected in advance and an- nounced in the group.

Meeting Minutes

Date: 19 September 2025
Time: 10:00 – 11:00
Venue: CM3

Meeting Details

Chairperson: Teboho Kele
Secretary: Nduvho Ramashia

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Jashna Gurahoo
- Promise Ndlovu
- Narsizo Macaringue
- Nduvho Ramashia
- Teboho Kele
- Rohan Gupta
- Abram Khoza
- Sibongiseni Jiyane
- Marcus Chitura
- Uwais Badat

Agenda

- Each group to present progress updates.
- Questions, clarifications, and advice after each presentation.
- Discussion on collaboration opportunities across groups.
- Planning for the next week and next meeting arrangements.

Project Progress Reports

Group 1: Distributed Resource Allocation for 5G Dense Cellular Networks Members: Nduvho Ramashia & Teboho Kele
The group has established base station and user numbers, selected a 5G numerology, and defined the influence structure. They are focusing on URLLC and eMBB services, mod- eling interference, and using machine learning for resource allocation (physical resource blocks and power). Initial random allocation is in place, with plans to implement ad- vanced algorithms. Contributions include improving spectrum efficiency and user fairness in dense networks, building on existing literature with a distributed approach.

Group 2: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane

The group is reviewing literature and defining system parameters for cell-free networks, focusing on resource sharing (power and physical resource blocks). Progress has been made in refining the problem formulation and considering baseline models, with particular attention to user association strategies. The group plans to improve upon existing studies using DRL and has identified opportunities for collaboration with Group 1 due to project similarities.

Group 3: AI-Based Modulation Recognition in Communication Systems Members: Promise Ndlovu & Narsizo Macaringue
The group has downloaded the RadioML 2018 dataset and uploaded it to a computing cluster for analysis under AWGN and Rayleigh fading channels. They have generated modulation types and scales, with next steps involving feeding the data into machine learning models. GPU challenges have been resolved, and VS Code is being used for remote cluster access.

Group 4: Energy Efficiency in IRS-Assisted Wireless Networks Members: Mohemad Kudia & Yasteel Ramphal
Developing energy-efficient aerial IRS systems with real-time reinforcement learning adaptation. Reported potential improvements of 45–60% in energy efficiency and 30–50% in spectral efficiency compared to baseline. Current work compares deep RL against tradi- tional optimization approaches.

Group 5: AI-Based Image Compression for Communication Systems Members: Rohan Gupta & Jashna Gurahoo
The group has implemented an autoencoder in Python to compress images from approximately 3MB to 0.5MB for transmission over Rayleigh fading channels. Challenges have been encountered in interfacing MATLAB and Python, as well as in sorting the DIV2K dataset for AI training. The group is currently exploring the implementation of Rayleigh fading channels in Python and adapting audio compression techniques for images. Future plans involve comparing the performance of these methods to traditional techniques.

Group 6: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul Members: Marcus Chitura & Uwais Badat
The group is focusing on edge computation offloading in 5G integrated access and back- haul (IAB) systems with mmWave wireless backhaul, aiming to optimize latency, device energy consumption, and system performance. They are reviewing literature to formulate their optimization problem and identify baseline studies, with a focus on resource alloca- tion strategies similar to those of Groups 1 and 2. Current efforts involve defining system parameters, such as the number of access points and users, and exploring simulation tools like MATLAB and Python.

Discussion and Questions

- Groups 1 & 2 identified overlaps in distributed vs. cell-free approaches and agreed to collaborate on literature review and optimisation methods.
- Group 3's cluster usage was noted as potentially useful for other teams facing computational bottlenecks.
- Group 4 discussed expected improvements in processing power, update frequency, and energy efficiency.

- Group 5 clarified compression goals, challenges in training, and relevance to satellite and mobile communication.
- Peer-to-peer support was encouraged, especially where projects share overlapping tools (e.g., ML models, HPC clusters).

Next Steps and Action Items

- Groups to finalise baseline papers and optimisation problem formulations this week.
- Collaboration to continue between Groups 1 & 2, and other teams with overlapping themes.
- Group 5 to investigate dataset preprocessing and Python channel implementation.
- Group 3 to begin deep learning model training on modulation dataset.

Next Meeting

- Venue: [To be Confirmed].
- Time: Friday 10:00.
- Chairperson and Secretary for the next meeting to be selected in advance.

Meeting Minutes

Date: 26 September 2025
Time: 10:00 – 11:00
Venue: EIE seminar room

Meeting Details

Chairperson: Promise Ndlovu
Secretary: Narsizo Macaringue

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Jashna Gurahoo
- Promise Ndlovu
- Narsizo Macaringue
- Nduvho Ramashia
- Teboho Kele
- Rohan Gupta
- Abram Khoza
- Sibongiseni Jiyane
- Marcus Chitura
- Uwais Badat

Agenda

- Each group to present progress updates.
- Questions, clarifications, and advice after each presentation.
- Discussion on collaboration opportunities across groups.
- Planning for the next week and next meeting arrangements.

Project Progress Reports

Group 1: Distributed Resource Allocation for 5G Dense Cellular Networks Members: Nduvho Ramashia & Teboho Kele
The group reviewed the literature and identified three key baseline papers. They are investigating efficient ways to allocate resources between heterogeneous traffic types, particularly balancing eMBB and URLLC requirements. They highlighted challenges in achieving user fairness under different network topologies and dynamic channel condi- tions. The group is exploring reinforcement learning methods, specifically Deep Rein- forcement Learning (DRL), to improve resource allocation strategies and overall system performance.

Group 2: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane

The group is working on channel estimation approaches for resource allocation in cell- free networks. They are considering both perfect and imperfect channel state information (CSI) and evaluating resource allocation strategies under these assumptions. Their study also examines centralized versus decentralized CSI frameworks, comparing the perfor- mance and complexity trade-offs of each approach.

Group 3: AI-Based Modulation Recognition in Communication Systems Members: Promise Ndlovu & Narsizo Macaringue

The group has completed dataset preparation using the RadioML 2018 dataset. Modu- lation schemes were verified, and data were uploaded to the Jaguar cluster with balanced sampling across SNR values (0–20 dB) and modulation types. The group generated scalograms and other signal representations for training. Initial CNN-based training has begun, with accuracy steadily improving over the first epochs. Next steps include ex- tending training to convergence, exploring hybrid deep learning models, and preparing for GUI integration.

Group 4: Energy Efficiency in IRS-Assisted Wireless Networks Members: Mohemad Kudia & Yasteel Ramphal

The group is developing a DRL-based framework for optimizing IRS-assisted communica- tion. Their algorithm evaluates multiple possible paths and selects the most optimal based on energy efficiency, spectral efficiency, robustness, and signal quality. Initial simulations modeled the IRS as a static object, while current work extends this to a drone-mounted IRS scenario, introducing trajectory and motion dynamics. They are implementing Soft Actor-Critic (SAC) reinforcement learning methods to improve adaptability and perfor- mance in dynamic environments.

Group 5: AI-Based Image Compression for Communication Systems Members: Rohan Gupta & Jashna Gurahoo

The group is developing a unique AI-based image compression model. They are moving beyond traditional classifiers, such as Random Forest, towards multi-layer approaches for classifying images into high- and low-quality categories. They reported a meeting with their supervisor for further guidance. Current efforts include transitioning from MATLAB to Python to better implement channel modeling, particularly Rayleigh fading, and refining their compression framework.

Group 6: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul Members: Marcus Chitura & Uwais Badat

The group is modeling an on-campus WiFi test scenario for edge computation with 5G integrated access and backhaul (IAB). They are exploring two routing approaches within a multi-objective optimization framework. The current setup considers 10 user equipment (UEs), 2 base stations, and 2 IAB nodes. Their work focuses on optimizing system performance and ensuring efficient resource allocation under the chosen architecture.

Discussion and Questions

- Group 3 was asked how their approach will compare with traditional methods. They reported that they have already run some simulations and obtained preliminary results on energy usage.

- Group 2 asked Group 4 about the implementation of the Soft Actor-Critic (SAC) method, requesting clarification on how it is being applied.

Next Meeting

- Venue: [To be Confirmed].
- Time: Friday 10:00.
- Chairperson and Secretary for the next meeting to be selected in advance.

Meeting Minutes

Date: 3 October 2025
Time: 10 am - 12 pm
Venue: CM3

Meeting Details

Chairperson: Mohemad Kudia **Secretary:** Yasteel Ramphal

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Marcus Chitura
- Uwais Badat
- Teboho Kele
- Nduvho Ramashia
- Sibongiseni Jiyane
- Abram Khoza
- Jashna Gurahoo
- Rohan Gupta
- Promise Ndlovu
- Narsizo Macaringue

Agenda

- Each group to present progress updates.
- Discussion of any issues encountered.
- Sharing of supervisor advice that may benefit other groups.

Project Progress Reports

Group 4: Energy Efficiency in IRS-Assisted Wireless Networks Members: Mohemad Kudia & Yasteel Ramphal
The group has successfully implemented a UAV aerial IRS system with basic trajectory parameters. Current work focuses on implementing more complex trajectory parameters to improve the system's performance. They are using deep reinforcement learning (DRL) to optimize energy efficiency in drone-based systems.

A key development is the implementation of additional basic machine learning solu- tions for comparison purposes. The supervisor emphasized the importance of demon- strating that their machine learning approach performs better than other ML solutions. Since the group is not using existing datasets or solutions from literature, they cannot directly compare their results to published research papers due to differences in scenarios and system structures. Therefore, they are implementing multiple baseline methods within their own simulation environment to ensure fair comparison.

The group noted that trajectory implementation is particularly challenging, as the updating needs to be very fast for drone applications. They estimate the project is nearing completion but acknowledged uncertainty regarding the difficulty of implementing the remaining trajectory features.

Supervisor Advice Shared:

- Compare "apples to apples" – don't compare a UAV system to a static system as they are fundamentally different.
- Don't compare results to published research unless using the same setup or configuration, as comparisons must be realistic.
- Justification is paramount – it's not about how much data you have or how great your results are, but rather your ability to explain why you made specific choices and what your results mean.
- Even poor results are acceptable if they can be properly explained, as the goal is understanding rather than achieving exceptional performance.

Group 1: Distributed Resource Allocation for 5G Dense Cellular Networks
Members: Nduvho Ramashia & Teboho Kele
The group has made substantial progress across all major components of their project. They have completed the system modeling, which includes the frame structure, numerol- ogy selection, bandwidth allocation, channel model, and resource model. The problem formulation is complete with a defined objective function and constraints including power budget per user connected to a base station, orthogonal multiplexing allocation, power order, and interference concerns.

Their solution framework uses deep reinforcement learning, specifically formulated as a multi-agent DRL problem with centralized training and distributed execution to match their distributed system architecture. They have defined the state space, action space, and reward function for their approach. Mathematical modeling and algorithm development are complete, and the system model is now implemented and functional.

The group is currently in the implementation and optimization phase, working to obtain results that demonstrate their algorithm performs better than other baseline al- gorithms. They are experiencing some challenges with unexpected results and are refor- mulating their optimization approach to achieve the desired outputs.

Novel Contributions: Building on their baseline paper, the group has added two key improvements to increase novelty:

- **User Mobility:** Unlike their baseline paper where users were static, they have implemented mobility to better mimic real-world scenarios where users move around with their devices while connected to the network.
- **NOMA (Non-Orthogonal Multiple Access):** Implemented to improve spec- trum efficiency by allowing two users to share the same spectrum.

Performance Metrics: The group is focusing on three key metrics: spectrum effi- ciency, user fairness (particularly for edge cell users who typically receive fewer resource allocations), and sum rate (network capacity). Their algorithm must demonstrate higher sum rate and better fairness compared to both their baseline paper and other traditional algorithms from literature.

Nduvho added that while their system may not necessarily perform significantly better than the baseline due to added complexity, the more realistic and complex system will still yield valuable results even with similar performance levels.

Group 2: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane
The group's project focuses on allocating resources in a cell-free wireless network. Similar to Group 1, they selected a baseline paper and aim to improve upon it by incorporating mobility into the system. In the baseline paper, the authors worked to improve spectral efficiency and minimize the number of links between users and access points (APs), which helps reduce fronthaul overload. Fronthaul overload refers to the communication between the CPU and access points, and minimizing it is essential for system optimization.

The group identified that while the baseline paper achieved some optimal results, it can be improved because it only considers stationary users. Their approach incorporates mobility to mimic real-life situations where users move at constant speeds. They are testing different topologies and varying scenarios to train their model to manage spectral efficiency based on user velocity. Each user's velocity is kept constant during testing.

Implementation Progress: The group has implemented their system in VS Code and is currently simulating the baseline paper approach to verify they can achieve similar results. Once validated, they will incorporate mobility and compare their results to both the baseline paper and other similar approaches in the literature.

Solution Framework: They are using a PPO (Proximal Policy Optimization) DRL algorithm for their multi-dimensional state space. Their action space involves activating and deactivating access points to minimize user-AP links, thereby reducing fronthaul load. The baseline paper's approach provides fairness but also results in starving more users. With mobility incorporated, only users that meet certain demand thresholds and are moving will be served by their strongest subset of access points.

Challenges: The group faced similar challenges to Group 1 in replicating baseline paper results due to incomplete information in the paper. They made logical assumptions for missing parameters and achieved results with similar interpretations, though not exact matches.

Novelty Concerns: When asked whether incorporating mobility alone is novel enough, the group explained that their supervisor suggested this approach and that mo- bility fundamentally changes the entire framework and problem formulation, significantly increasing complexity. They plan to meet with their supervisor to confirm the novelty is sufficient.

Group 5: AI-Based Image Compression for Communication
Systems Members: Rohan Gupta & Jashna Gurahoo
The group reported a challenging week but made meaningful progress. Following their meeting with their supervisor last Friday, they received approval to proceed with their classification algorithm approach.

Classification Approach: They are using a Random Forest classifier to identify image types, which then determines the appropriate AI compression algorithm to ap- ply. This week, they developed a workaround for their dataset classification problem by modifying their Python code to automatically analyze images and determine their type, eliminating the need for manual classification.

Feature Extraction: The image analyzer extracts several features including edge density, texture complexity, colour variance, brightness mean, contrast measure, and entropy. Based on these features, the Random Forest classifier categorizes images (e.g., high quality, low complexity, natural scene) and identifies the appropriate compression method.

Next Steps: The group has identified a group of compression algorithms for potential use and expects to finalize the specific algorithms by next week. They are currently testing whether direct AI training is necessary or if the AI can implement compression without explicit training. This decision will be determined by performance evaluation by the end of next week.

Group 3: AI-Based Modulation Recognition in Communication Systems Members: Promise Ndlovu & Narsizo Macaringue
The group is building an automatic modulation recognition system. They selected a baseline paper that used a CNN (Convolutional Neural Network) approach and success- fully replicated the baseline results, achieving approximately 80% accuracy similar to the original paper.

Improvements and Results: After implementing their improvements to the CNN approach, they achieved 92% accuracy with 10 modulation types. However, some modu- lation schemes still require tweaking and improvement.

Novel Contributions: To improve upon the baseline CNN approach, they added two key methods:

- **DRSN (Deep Residual Shrinkage Network):** Suppresses noise in the modu- lations.
- **GCN (Graph Convolutional Network):** Learns the spatial features of the dataset, identifies relationships, and propagates this information to build a more comprehensive model.

Common Challenge Addressed: Promise noted that their improved model aims to address a common trend in many implementations where models struggle to distinguish between two similar modulation schemes. This is a recognized problem across multiple papers in the field, and their project specifically targets this issue.

Infrastructure Change: The group moved from using VS Code to Google Colab due to problems they encountered with VS Code.

Group 6: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul Members: Marcus Chitura & Uwais Badat
The project focuses on millimeter wave 5G networks with edge computation. Since their last update, the focus has been on scaling up the system to make it more realistic by incorporating path losses and noise into the simulation, as recommended by their super- visor.

Progress: Implementation has been progressing well. Before reaching their current scale of 70 user equipment, they have been achieving good efficiency and acceptable latency values. Marcus indicated the work is going well overall.

Discussion and Questions

Questions for Group 1: Sibongiseni Jiyane (speaking on behalf of Abram Khoza from Group 2) asked several clarifying questions about what specific results the group hopes to demonstrate, whether comparisons would be made strictly to the baseline paper or also to other papers solving similar problems, and inquired about the mobility implementation and its constraints.

Teboho confirmed they would compare against both their baseline paper (to show novelty through improvements) and other traditional algorithms from different papers to demonstrate overall performance superiority. He clarified that mobility does not directly relate to their constraints but serves to make the system more realistic and complex, addressing graduate attribute requirements for complexity.

Questions for Group 2: Teboho asked Group 2 about their implementation chal- lenges with the baseline paper, their solution algorithm choice (PPO DRL), and whether incorporating mobility alone provides sufficient novelty. The groups had similar expe- riences with incomplete baseline paper information and agreed to continue discussions after the meeting.

Computing Cluster Access Issues: Sibongiseni Jiyane reported ongoing issues with storage limitations on the computing cluster (specifically the class.gpu node). He encountered missing libraries (such as torch) and storage complaints when trying to create a virtual environment for his project. He had contacted Professor Nixon, who indicated the issue is outside the school's control and would get back to him.

Marcus Chitura and Teboho Kele advised speaking with Ken Nixon about permissions or custom configurations needed on the cluster.

Mohemad Kudia suggested that the storage limitations might be related to the number of concurrent users accessing the system, referencing an earlier mention that usage would be limited by the number of simultaneous users.

Sibongiseni also inquired whether it is possible to run code on the cluster without setting up a virtual environment, noting that some required libraries appeared to be already installed globally, though storage constraints still prevented execution. Mohemad indicated he did not know the answer, as he has been able to run his project on his local computer and has not needed to use the cluster.

Administrative Matters

Meeting Minutes Documentation: At the meeting's conclusion, Mohemad committed to sending out the meeting minutes as soon as possible. He requested that someone share the link to the LaTeX document used for compiling meeting minutes, as access to the collaborative document needs to be confirmed. Promise Ndlovu agreed to send the link.

Next Meeting

- Venue: [To be Confirmed]
- Time: Friday before 12:00
- Chairperson and Secretary for the next meeting to be selected in advance.

Meeting Minutes

Date: 10 October 2025

Time: 10:30 – 11:15

Venue: CM3

Meeting Details

Chairperson: Jashna Gurahoo

Secretary: Rohan Gupta

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Jashna Gurahoo
- Promise Ndlovu
- Narsizo Macaringue
- Nduvho Ramashia
- Teboho Kele
- Rohan Gupta
- Abram Khoza
- Sibongiseni Jiyane
- Marcus Chitura
- Uwais Badat

Agenda

Each group reported on their weekly findings.

Project Discussion

Group 1: AI-Based Image Compression for Communication Systems

Members: Rohan Gupta & Jashna Gurahoo

The group reported that they have progressed beyond data classification, having developed code to automatically classify their image dataset through feature extraction techniques. These features identify image types, enabling the application of their data for compression. They confirmed that the developmental stage of their AI training model has been completed, with current efforts focused on optimising the compression performance through further model training. Work has also commenced on both the API and GUI components — the API links the data and Python-based compression process to the front-end GUI, creating a visual and interactive presentation layer. For the coming week, the group intends to evaluate the performance metrics of their model and benchmark results against existing academic models, drawing data and performance parameters from published research for comparison.

Group 2: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane

The group reported progress in replicating the baseline paper identified during their literature review. They successfully reproduced the initial results and have since begun extending the model by incorporating mobility into the system design. However, they noted that their initial approach to modelling mobility was not implemented correctly and will therefore be revised. The group plans to revisit relevant literature to ensure accurate integration of mobility into their framework. Their overall objective remains focused on optimising spectral efficiency within the cell-free wireless network. Once the revised mobility model has been correctly implemented, they intend to conduct hyperparameter tuning to further investigate and improve model performance.

Group 3: Distributed Resource Allocation for 5G Dense Cellular Networks

Members: Nduvho Ramashia & Teboho Kele

The group reported ongoing efforts to replicate the results of their baseline paper to enable performance comparison between the reference model and their own implementation. This process has presented challenges, primarily due to incomplete or inconsistent information within the paper—specifically, discrepancies between the described methodology and parameter tables, which have made accurate replication difficult. Following recent consultation with their supervisor, the group has decided that full replication may not be necessary. Instead, they will use the published results and conclusions from the baseline study as a benchmark for comparison. Their focus for the upcoming week is to evaluate and compare their model's performance against these reported findings to validate the system's effectiveness.

Group 4: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul

Members: Marcus Chitura & Uwais Badat

The group reported progress on implementing a computation offloading algorithm designed to enhance user equipment energy efficiency and ensure timely task execution within a 5G integrated access and backhaul (IAB) system. The focus remains on evaluating whether tasks offloaded to the network can meet latency and deadline requirements across different network slices, depending on task size. They confirmed that the simulation environment has been successfully established, and benchmarking has been completed against two existing algorithms. The results obtained so far indicate satisfactory system performance. The group plans to include a third benchmark algorithm in the coming week to strengthen the comparative analysis and further assess overall network performance.

Group 5: AI-Based Modulation Recognition in Communication Systems

Members: Promise Ndlovu & Narsizo Macaringue

The group reported that they successfully developed and trained their model and completed the initial GUI design. During the week, they presented their results to the supervisor, who provided constructive feedback and highlighted areas requiring improvement. Specifically, the supervisor identified inconsistencies in the model results and requested greater transparency within the GUI to ensure that each step of the modulation recognition process is clearly visible to users. This will help verify that the model's outcomes are generated purely by the system without external input or bias. The group's focus for the coming week will be on enhancing the GUI interface for better transparency and refining the model's performance to address the identified issues and ensure accurate, verifiable results.

Group 6: Energy Efficiency in IRS-Assisted Wireless Networks
Members: Mohemad Kudia & Yasteel Ramphal

The group reported that the design phase of their project is complete, and they are now preparing to conduct extensive training to evaluate the model's performance. Following feedback from their supervisor, they considered implementing a cascaded IRS configuration; however, they noted that this approach introduces numerous additional parameters and would make a simple proof-of-concept simulation impractical.

Furthermore, such an extension would diverge from the project's primary focus on energy efficiency, and therefore, the group plans to maintain its original scope while refining the current design and conducting detailed performance analysis.

Discussion on Inter-Team Collaboration

After the main session, groups with overlapping project themes remained to discuss shared aspects of their work. They exchanged thoughts, opinions, and approaches on similar topics, fostering collaboration and mutual understanding. These informal discussions provided an opportunity to share insights, compare methodologies, and explore possible areas for support between teams.

Next Meeting

- Venue: [To be Confirmed]
- Time: Friday before 12:00
- Chairperson and Secretary for the next meeting are Marcus Chitura and Uwais Badat.

Meeting Minutes

Date: 17 October 2025

Time: 10:30 – 11:15

Venue: CM3

Meeting Details

Chairperson: Uwais Badat

Secretary: Marcus Chitura

Attendees

- Mohemad Kudia
- Yasteel Ramphal
- Jashna Gurahoo
- Promise Ndlovu
- Narsizo Macaringue
- Nduvho Ramashia
- Teboho Kele
- Rohan Gupta
- Abram Khoza
- Sibongiseni Jiyane
- Marcus Chitura
- Uwais Badat

Agenda

Each group reported on their weekly findings.

Project Discussion

Group 1: AI-Based Image Compression for Communication Systems

Members: Rohan Gupta & Jashna Gurahoo

The group noted that they have made significant progress towards gathering performance metrics of the AI training model with a desire to benchmark against existing models so as to gather useful data that can either push for further adjustments of the existing model or finalizing of the results if the collected results are good.

Group 2: Resource Allocation in Cell-Free Wireless Networks
Members: Abram Khoza & Sibongiseni Jiyane

The group managed to revisit their initial model for mobility, making it more accurate to the model within the baseline paper, ensuring a fair comparison. A hyperparameter tuning was being implemented by the group to further investigate and improve model performance. Steady progress was being made to achieve and the group hopes this process is a smooth one, as the investigation is coming towards a end.

Group 3: Distributed Resource Allocation for 5G Dense Cellular Networks

Members: Nduvho Ramashia & Teboho Kele

The group approached an evaluation of their model to compare with a restrained model of the baseline paper as, the full model could not be formed due to complications with a inconsistent and incomplete information with the paper. The group managed to collect results and will move towards evaluation of these results hoping that no further modifications need to be made as time is running out.

Group 4: Edge Computation for a 5G IAB System with mmWave Wireless Backhaul

Members: Marcus Chitura & Uwais Badat

The group reported progress on implementing a computation offloading algorithm designed to enhance user equipment energy efficiency and ensure timely task execution within a 5G integrated access and backhaul (IAB) system. The latest progress reported by the group is the successful implementation of the benchmark with the proposed algorithm against three other algorithms. The results were sent to the supervisor for confirmation of quality, in the next week the group hopes to prepare in time for the Open Day Poster.

Group 5: AI-Based Modulation Recognition in Communication Systems

Members: Promise Ndlovu & Narsizo Macaringue

The group reported that they successfully developed and trained their model and completed the initial GUI design. The group implemented the suggested modelling from the supervisor, in regards to removing modelling inconsistencies and providing a greater transparency modelling approach within the GUI to ensure that modulation recognition is visible to the users. The group reported good progress in regards to this endeavor, and hope to verify the new modelling angle by running it past their supervisor. The group hopes to get feedback in time to make the Open Day poster for presentation.

Group 6: Energy Efficiency in IRS-Assisted Wireless Networks
Members: Mohemad Kudia & Yasteel Ramphal

The group continued to pursue the original intent of the project to focus on energy efficiency at the behest of the comment by the supervisor, who had suggested a push for implementing a cascaded

IRS configuration. The group reported good results captured with the original, method and will proceed to prepare for the Open Day presentation with the collected results.

Attempted implementing cascaded ris but failed due to too many parameters factoring in as well as increased computational costs. Proof of concept simulations would be too unrealistic. Implemented a live dashboard for presentation as well as 3D plots for presentation. Researched edge computing for reducing computational load on uavs for future work.

Discussion on Inter-Team Collaboration

After the main session, groups with overlapping project themes remained to discuss shared aspects of their work. They exchanged thoughts, opinions, and approaches on similar topics, fostering collaboration and mutual understanding. These informal discussions provided an opportunity to share insights, compare methodologies, and explore possible areas for support between teams.

Next Meeting

- Venue: [To be Confirmed]
- Time: Friday before 12:00
- Chairperson and Secretary for the next meeting are not yet determined.